Brain Behav. Evol. *14:* 385–398 (1977)

Comparative Investigation of the Relationship between Cerebral Indices and Learning Abilities

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Key Words. Cerebral indices · Comparative psychology · Comparative learning · Discrimination learning

Abstract. Indices of cerebral development were computed for 23 different species in order to assess their capacity to accurately reflect differences in learning ability. The resulting correlations between index values and performance on a variety of tasks strongly suggest that this approach may offer the best type of continuum for the comparative study of learning. It was further suggested that if the index proposed by Jerison were expanded to reflect possible neuronal connections, a more powerful measure might be obtained.

While recent critics have written an epitaph for comparative psychology [20], noted it lacked an appropriate theoretical base [11], and questioned the generality of 'laws of learning' [34], the basic problems could be largely solved if a valid comparative continuum were available, and if a behavioral measure which was independent of species-specific abilities could be found. Although the problem with the adequacy of the dependent variable is not amenable to simple solutions, the data presented by RIDDELL *et al.* [29] suggest that it may be possible to factor out species-specific abilities from error scores, thus obtaining a purer measure of intrinsic learning abilities. The investigation of the relationship between learning abilities and measures of cerebral development has been limited to a few studies [21, 27–29], and no formal attempts have been made to either choose between measures or to apply these measures to the wealth of behavioral data previously collected. The basic purpose of this paper is therefore twofold: first, to ascertain the success of cerebral measures in

[1] We would like to thank FRED GRAVETTER and SACHIO ASHIDA for critical readings of earlier drafts of the paper and STAN WECHKIN for his intellectual support.

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predicting learning ability over a wide range of tasks; and second, to evaluate those factors within the indices responsible for these predictions.

DUBOIS [5] developed the first index of cephalization based on SNELL's [36] observation that the relation of the brain size to the body size follows the general allometric function $E = kP^{a}$, where E and P are the weights or volumes of the brain and body, respectively, a is a constant, and k varies across species. Since then, numerous other indices have been proposed, e.g., WIRZ's Scaled Ratio [43] and SACHER's Species Factor 1 [33]. Although the majority of indices have been based on the allometric relationship, considerable refinement has been necessary to develop the values for k and a. JERISON [17] and STEPHAN and ANDY [37] have confirmed the view that $a = 0.66$. The value for k, however, has varied according to the purpose of the specific index, and is generally regarded as an index of cephalization [5, 17, 37] which has been used primarily to compare orders within the mammalian lineage. WIRZ's *Totalindex* was developed by dividing the E for a contemporary mammal by a value representing that portion of a mammalian brain specifically limited to bodily or 'primitive' functions. This primitive portion was termed the *Stammreste*. WIRZ considered the living insectivores as the most primitive of living placentals and defined the *Stammreste* as corresponding to the size of a brain stem complex of such a living insectivore of the same body weight. While STEPHAN and ANDY [37] also developed an index on the insectivores, they referenced the brain weight of a species to the total brain weight of certain living insectivores considered 'basal' due to their limited cortical development, rather than to the brain stem alone. The development of the reference value in this manner was based on the assumption that placental animals are thought to originate from primitive insectivores and that certain living insectivores can be considered relic forms. JERISON [15] broadened the concept of a cephalization index by suggesting that such an index may be used to indicate brain evolution in mammals, and further postulated that the coefficient is related to behavioral differences. A similar view has been advanced by HERSCHEL [10] and SACHER [33]. Studies by RIDDELL *et al.* [29] and MASTERTON and SKEEN [21] using cerebral indices as the independent variable have provided some support for this view.

While many cerebral indices are available, only those suggested by HERSCHEL [10], JERISON [17] and RENSCH [26] were critically evaluated here for the purpose of assessing their predictive success. These three measures were chosen in an attempt not only to determine the predictive

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power of this approach, but also to ascertain the power of the different approaches in indicating those factors which affect learning abilities. JERISON's index was chosen since it is not only the most highly developed of the indices but it also correlates highly with other measures [33, 37]. The measure suggested by HERSCHEL represents a simple approach to the problem of brain/body relationships by judging all species relative to the standard cube/square function. Since RENSCH's index, brain size *per se*, represents the simplest approach of all, comparing these three measures with each other would determine if additional power in assessing cerebral development is generated by the addition of parameters.

HERSCHEL [10] citing KROEBER [19] noted that the relationship of brain size to body size in mammals is analogous to the relationship of the surface area to the volume of the body, with body weight increasing as a cubic function while the brain weight increases as a squared function as one goes from small to large animals. In an attempt to isolate differences in brain size then, he converted the absolute data into a scaled ratio that accounts for the different exponential rates of change. The resulting index, $\sqrt[3]{B}/\sqrt{b}$, where B equals body weight and b equals brain weight, reflects any resultant disproportion between species. Since the data that he presented correlated roughly with behavioral differences between species, HERSCHEL suggested that a physiological index such as this may, in general, covary with the observed species differences. Finally, HERSCHEL maintained that this index correlated only with behavior and that it was independent of body weight.

The evolution of JERISON's 'Extra Neurons' index (N_e) may be traced through several articles [13, 15] and is presented fully in his 1975 book. In general, this formulation is based on the following lines of reasoning: (1) for archaic mammals there is a general relation between E and P given by the allometric equation $E = 0.03P^{2/3}$; (2) E can be subdivided into two parts, E_v, the portion of E associated with primitive functioning, and E_e, the portion of E associated with intelligence; (3) in primitive mammals $E = E_v$; (4) the equation $E_v = 0.03P^{2/3}$ can be employed to calculate that portion of E in contemporary mammals allocated to primitive function; (5) E_e may then be calculated by subtracting E_v from E, and (6) E_e is then converted into total neurons available for other than primitive functioning, the resulting number of neurons being designated N_e. N_e thereby indicates the number of neurons in that portion of the brain defined as E_e, and is a numerical measure of progressiveness beyond that required for a given body size.

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In his 1966 article, HOLLOWAY [12] criticized the type of index proposed by both HERSCHEL [10] and JERISON [15] on the grounds that they did not consider such factors as neural/glial interaction or differential dendritic branching. Since detailed information relative to the first factor is not available, only the second will be treated here. As RENSCH [26] and JERISON [17] have noted, an increase in the absolute size of the brain is generally paralleled by an increase in the size and number of cortical neurons and correspondingly more dendritic interconnections. These authors suggested that this results in greater configurational complexity and hence possibly greater behavioral plasticity. RENSCH [26] in citing behavioral evidence for species as divergent as trout, bats, and rats hypothesized that brain size (E) *per se* is the most appropriate index. This was based largely on the view that the most important variable is the number of possible neural connections and the richness of individual neurons. Since both factors increase with increases in E, E itself is then the critical consideration. There is some independent evidence to support this notion. CLENDINNEN and EAYRS [3] reported that the offspring of rats treated with anterior pituitary growth hormone made fewer errors per problem and could solve more difficult problems than the controls. HOLLOWAY [12] noted that experience may modify the number of dendritic interconnections and that this may be reflected in changes in behavioral capacity. A similar finding is reported by ROSENZWEIG *et al.* [32].

It follows from the preceding discussion that some measure that would be representative of neural interconnections would be advantageous. As can be seen, however, the variables affecting such a measure are too complex to permit a precise estimate of the actual number at this time. It is for these reasons that it is proposed that N_c be extended to $N_c^2/_2$. Using JERISON's N_c as a base, this provides a numerical value that reflects the maximum possible dendritic interconnections for a given species, assuming the possibility of one and only one interconnection between any two given neurons. The parameter N_c was chosen rather than N since the use of N_c allows one to factor out of E that portion of E that has been shown to reflect behavioral differences rather than differences in P – e.g., CLENDINNEN and EAYRS [3] reported that increases in dendritic interconnections are reflected in mediated behavior only; not primitive or reflexive behavior. This transformation of N_c then tries to account for the increase in permutation possibilities which occur with increases in E, while controlling for differences in P. While it is not suggested that this formulation gives an absolute count of actual neuronal interconnections, it does

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indicate relative differences between species. If, in the future, a more accurate count seems desirable, a formulation similar to that suggested by UTTLEY [39] might be employed, given the appropriate morphological data.

Since a crucial test of any index is the degree of correlation between that index and behavioral plasticity, each of the brain indices (BIs) discussed were evaluated in terms of their correlation with tasks shown to be sensitive to species differences. The morphological and behavioral data were taken from previously published studies.

Results

Using the data presented by VON BONIN [40], Pearson correlations involving all possible combinations of brain weight, body weight and the indices were calculated, and the results are presented in table I. As may be seen, though H and N_c are significantly correlated with body weight apparently violating the claims for each index, the values represent only 9 and 5% of the total variation, respectively. Secondly, the high correlation between body weight and brain weight is to be expected in light of the previously demonstrated allometric relationship. Finally, given the assumption that all of the indices tap the same type of capabilities, one

Table I. Correlation matrix for anatomical data and brain indices

Variable		r
E	P	0.805**
E	N	0.981**
E	N_c	0.749**
E	H	0.156
P	N	0.728**
P	N_c	0.235*
P	H	0.332**
N	N_c	0.831**
N	H	0.070
N_c	H	−0.223*

E = Brain weight, P = body weight, N = number of cortical neurons calculated by formula $8 \times 10^7 E^{2/3}$, N_c = Jerison's index, and H = Herschel's index. * $p<0.05$, ** $p<0.01$.

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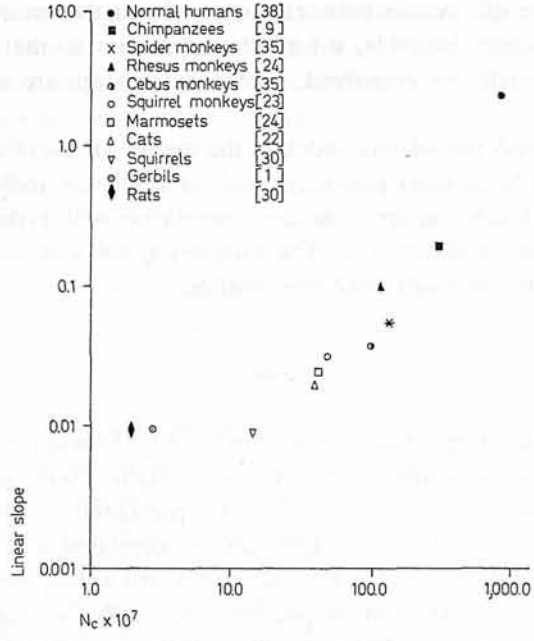


Fig. 1. Linear slope of learning set performance of 11 species as a function of N_c. Figures in brackets denote references.

would expect high correlations among them. The correlation between H and N_c, although in the right direction, was insignificant, a result which suggests that those two indices may well be assessing different sources of variance.

Behavioral Correlations

The BIs and performance data for a total of 23 species are listed in table II on five separate tasks: learning sets (LS), successive discrimination reversals (SDR), delayed alternation (DA), conditioned response reversals (CRR), and extradimensional shifts (ED). The element common to all of these behavioral measures appears to be that of inhibition, which also seems to be a common denominator for those tasks which have demonstrated consistent species differences [27]. Since it seemed presumptuous to assume that the BI and performance scores lay on equal interval scales, they were ranked and the resulting Spearman rho's are presented in table III.

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Table II. Anatomical and performance data for several species on five tasks

	Species	Dependent variable	P, g	E, g	$N_e \times 10^7$	H	Slope	Behavior rank	N_e rank	E rank	H rank	Behavior reference	E and P reference
1	*Cebus capuchin*	successive discrimination reversal	3,765.0	73.8	110.0	5.72	–	1	1	1	3	8	17
2	*Procyon lotor*		4,290.0	39.2	60.4	8.2	–	2	2	2	5	8	4
3	*Mephitis mephitis*		1,980.0	10.2	15.0	12.5	–	3	6	6	6	8	4
4	*Nasua nasua*		399.0	23.3	54.1	4.8	–	4	3	5	1	8	4
5	*Saimiri sciurea*		630.0	24.8	54.0	5.44	–	5	4	4	2	8	17
6	*Potos flavus*		2,620.0	31.1	53.0	7.8	–	6	5	3	4	8	4
1	*Pan troglodytes*	conditioned response reversal	50,340.0	382.5	325.6	6.0	–	1	1	1	1.5	41	18
2	*Papio hamadryas*		16,000.0	179.0	197.0	6.0	–	2	2	2	1.5	41	17
3	*Canis familiaris*		13,520.0	80.6	96.2	8.4	–	3	3	3	3	41	4
4	*Silvilagus gabbi*		539.0	4.5	10.2	11.3	–	4	4	4	4	41	4
1	*Galago-senegalensis*	delayed alternation	200.0	5.0	15.2	8.3	–	1	1	1	1	21	4
2	*Tupaia glis*		145.5	3.2	10.1	9.4	–	2	2	3	2	21	27
3	*Erinaceus europacus*		860.0	3.35	2.39	16.4	–	3	3	2	3	21	17
1	*Homo sapiens*	learning sets	64,807.4	1,326.6	855.6	3.47	2.5	1	1	1	1	38	4
2	*Pan troglodytes*		28,022.9	357.1	329.8	4.97	0.20	2	2	2	3	9	18
3	*Macaca mulatta*		8,719.0	106.4	136.0	6.3	0.11	3	4	4	6	23	17
4	*Ateles geoffroyi*		7,787.0	117.0	150.0	5.79	0.06	4	3	3	5	35	17
5	*Cebus albifrons*		1,640.0	80.0	128.0	4.17	0.04	5	5	5	2	35	17
6	*Saimiri sciurea*		630.0	24.8	54.0	5.44	0.03	6.5	6	7	4	24, 35	17
7	*Callithrix jacchus*		850.0	22.0	47.2	6.38	0.03	6.5	7	8	7	24	10
8	*Felis domesticus*		3,275.6	25.3	40.7	9.33	0.02	8	8	6	8	22	2
9	*Meriones unguiculates*		62.0	1.0	3.0	12.70	0.01	10	10	11	10	1	present study
10	*Rattus rattus*		326.6	1.9	2.4	15.55	0.01	10	11	10	11	42	present study
11	*Sciurus vulgaric*		389.0	6.0	15.5	9.42	0.01	10	9	9	9	30	40

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Table II (continued)

	Species	Dependent variable	P, g	E, g	$N_e \times 10^7$ H	Slope	Behavior rank	N_e rank	E rank	H rank	Behavior reference	E and P reference
1	*Homo sapiens*	extradimensional shift	64,807.4	1,326.6	855.6	3.47	–	1	1	1	1	29	4
2	*Cebus albifrons*		1,640.0	80.0	128.0	4.17	–	2	2	2	2	29	17
3	*Saimiri sciurea*		630.0	24.8	54.0	5.44	–	3	3	4	3	29	17
4	*Paca paca*		6,100.0	28.7	37.9	10.83	–	4	4	3	5	29	29
5	*Tupaia glis*		145.4	3.2	10.1	9.4	–	6	5	5	4	29	27
6	*Rattus rattus*		326.6	1.9	2.4	15.6	–	5	6	6	6	29	28

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Table III. Spearman correlations between brain indices and performance

Task	N_c	H	E
Learning sets	0.98	0.87	0.95
Successive discrimination reversals	0.66	−0.26	0.43
Conditioned response reversals	1.00	0.95	1.00
Delayed alternation	1.00	1.00	0.50
Extradimensional shifts	0.94	0.92	0.88
$\bar{p}\mu$ mean Spearman correlation (weighted by d.f.)	0.92	0.66	0.82
$\bar{e}^2w$	0.85	0.44	0.68

Learning Sets

Since learning rates have been previously mentioned as appropriate measures for species comparisons [24, 35], the slope of LS curves from chance to asymptotic performance was used. The slopes were calculated from published data, with the hypothetical starting point of 0 trials, 50% correct trial 2 responses; or chance. It should be noted that although SHELL and RIOPELLE [35] used different procedures for their squirrel monkeys than did MILES [23], the slopes when rounded to two significant figures, are identical for both studies.

The failure to obtain a perfect correlation between N_c and performance is a function of both the lack of performance differences in Rodentia and the inversion of spider and rhesus monkeys. The Rodentia data are probably best interpreted by assuming that there is a basement effect on this task at N_c values lower than approximately 12×10^7. This conclusion is based on ROLLIN's [30] observation that after 1,000 problems the performance of rats and gerbils is extremely poor (less than 60% correct of trial 2) and no differences are evident between groups. The inversion of spider and rhesus monkeys might well be function of sampling error. The N_c values for these species differed by only 14×10^7, i.e., approximately 10% of the index values for the species. Because intraspecies variation of the same relative magnitude has been found for rats and gerbils, it should not be assumed that values this similar actually represent distinct distributions. Since confidence levels around many of the indices are not available however, no attempt was made in either this analysis or subsequent correlations to take into account the possibility of the lack of significant differences between index values or behavioral measures.

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Successive Discrimination Reversals

The measure used for ranking SDR performance was mean total errors across 19 reversals. The results of these rankings are shown in table II and BI × behavioral measure correlations may be seen in table III. The failure to find a perfect correlation between performance and N_c is due to the reversal performance of the skunk which is considerably better than that which would be predicted by the index.

Conditioned Response Reversals

This analysis employed the trials to criterion scores given by VORONIN [41]. As is evident from table III each of the BIs correlate highly with performance on this problem.

Delayed Alternation

The measure used for this analysis was total errors. The lack of a significant correlation for H was due to lack of H value differences between the bush baby and the tree shrew. It should be noted that delayed response performance as reported by MASTERTON and SKEEN [21] would have yielded the same correlations for each of the indices.

Extradimensional Shifts

As is evident from table III, all indices correlated highly with performance over a series of ED shifts. The behavioral measure employed for the analysis was K as developed by RIDDELL *et al.* [29], and is basically a factor which is derived from the solutions of simultaneous equations whereby performance is assessed after problem difficulty and species-specific abilities are taken into account.

Discussion

It was evident that while all BIs generally correlated positively with learning, JERISON's N_c was consistently the best predictor. The average correlation of all five learning tasks with N_c, computed by weighing each of the correlations by its degrees of freedom, was equal to 0.92, indicating that this brain index is highly successful at predicting the ordering of species on learning tasks. Similarly, when these mean correlation coefficients are squared, it may be seen that while all BIs share common variance with the learning measure, it is greatest for N_c. The superiority of brain size alone (E) over H is interesting since, within the limited range of

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species sampled, it indicates that the possible number of interneuronal connections, regardless of body weight (P) may be more important than deviations from the square/cube relationship between E and P. Since N_c is a measure of E after accounting for that part E associated with P (E_v), its added predictive power is due to the ability to factor out the effect of P on E.

The differences in correlations across tasks are probably attributable to both the range and number of species sampled and the susceptibility of the tasks to species-specific abilities. Reversal learning, for example, varies within species as a function of task difficulty [7], and may also be affected by motivational variables [6]. In such cases it is likely that the low correlations are a function of the dependent variable rather than the inadequacy of the brain index approach. Similarly, those tasks which show the most consistent correlations across indices, i.e., learning set and extradimensional shifts, employ dependent variables which are designed to minimize species differences.

Conclusion

While the brain indices employed in the present study do not represent nearly the last word in neuronal/behavioral relationships, it is clear that reliable relationships do exist between brain indices and learning ability. Further investigation for the purpose of determining which indices are the best would be useful, as well as further refinement to take into account the criticisms that have been offered.

One of the main criticisms of this approach is that it does not take into consideration differential cerebral organization. However, as JERISON [17] has noted, it may be that the differences in neuronal configuration across species are to be expected for the same reason that behavioral specialization exist, i.e., as adaptations to niches. Placing the major emphasis on these specializations will not lead to the discovery of ordered differences among species, whereas an approach based on a more generalized notion such as a brain index continuum may be much more productive in this respect.

JERISON [17] has recently provided evidence that, at least within the mammalian lineage, the total information that can be processed by a given mass of cortical tissue is constant across species; any difference in information processing ability across species, therefore, must be a function of different amounts of cortex. This relationship is supported by the find-

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ing that different amounts of cortex are correlated with the number of neurons therein, and, therefore to the number of possible interconnections. It is for these reasons as well as the positive relationship between BIs and learning reported herein that it is proposed here that brain indices presently offer the best possible basis for comparative studies of learning abilities.

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